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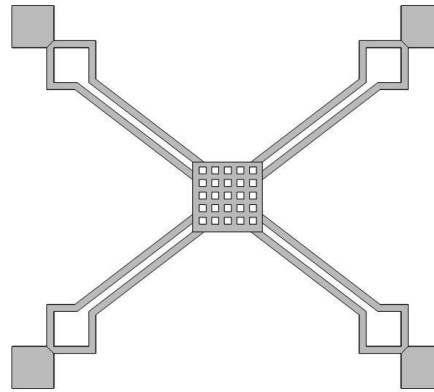
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Thalavapalayam, Karur - 639 113, TAMILNADU.



Design and Analysis of MEMS Accelerometer for Space Launch Vehicle Application



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Problem Statement

Space launch vehicles experience extreme vibration, shock, and acceleration during launch operations, which require highly accurate and reliable acceleration measurement. Conventional accelerometers are often bulky, consume higher power, and may fail under harsh aerospace conditions. Therefore, there is a need for a compact, high-sensitivity MEMS-based accelerometer capable of operating reliably in space launch environments.



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Literature Review

MEMS accelerometers are widely used to measure acceleration and vibration in aerospace systems, especially in space launch vehicles. Traditional accelerometers are often bulky, consume more power, and may not provide the required sensitivity and reliability under extreme launch conditions such as high vibration, shock, and temperature variations. These limitations make it difficult to achieve accurate real-time motion sensing during launch operations.

Although advanced accelerometer technologies exist, many are not optimized specifically for space launch vehicle environments. They may lack proper structural optimization, miniaturization, and performance analysis required for high-precision aerospace applications.

Limitations Identified

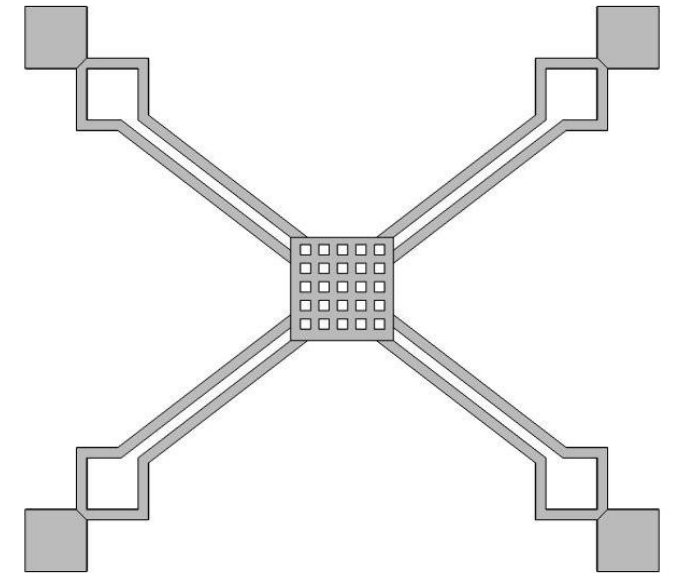
Existing accelerometer designs do not fully address the need for compact size, high sensitivity, low power consumption, and robust performance under harsh launch conditions. This creates the need for a carefully designed and analyzed MEMS accelerometer that ensures accuracy, reliability, and suitability for space launch vehicle applications.



Project Overview

This project presents a MEMS-based accelerometer designed for space launch vehicle applications, capable of measuring acceleration through proof mass displacement.

- High-sensitivity MEMS structure for harsh aerospace environments.
- Compact, low-power design with improved mechanical stability.
- Symmetric proof-mass design ensures uniform stress distribution and reliable sensing performance.
- Suitable for launch vehicle guidance, vibration monitoring, and aerospace inertial sensing applications.





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Objectives

- To design a MEMS-based accelerometer suitable for space launch vehicle applications.
- To develop a centrally suspended proof mass structure for high-sensitivity acceleration measurement.
- To achieve uniform stress distribution using symmetric suspension beam design.
- To analyze displacement and stress behavior using COMSOL Multiphysics simulation.
- To enhance mechanical stability and reliability under high vibration and shock conditions.
- To develop a compact, low-power accelerometer for aerospace navigation and structural monitoring.



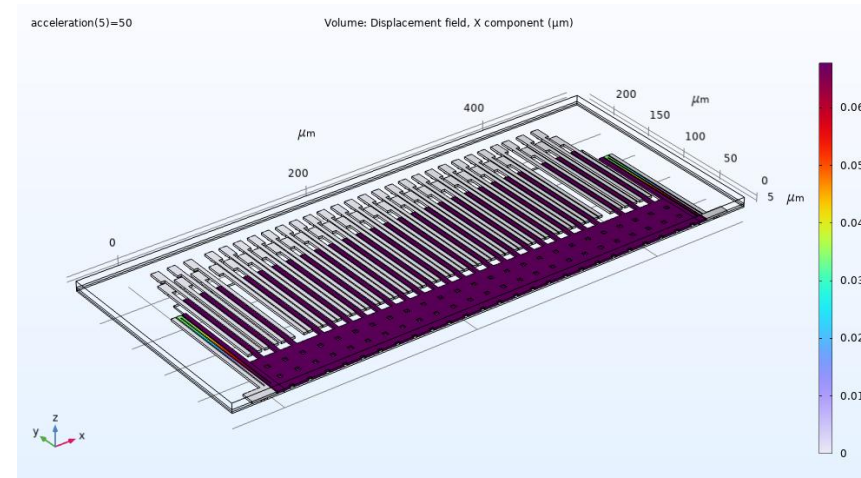
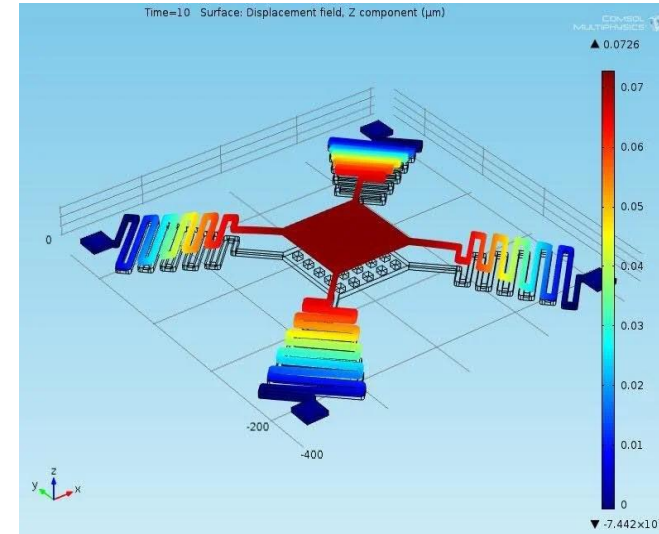
Existing vs Proposed System

Existing System

- Uses conventional accelerometers (bulky)
- Limited sensitivity under extreme conditions
- Higher power consumption
- Poor performance in high vibration & temperature
- Less structural optimization

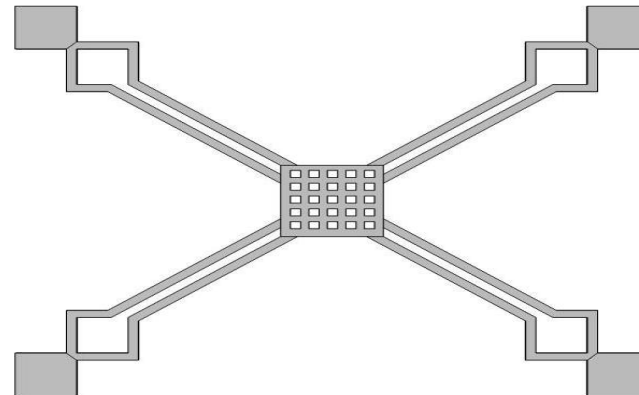
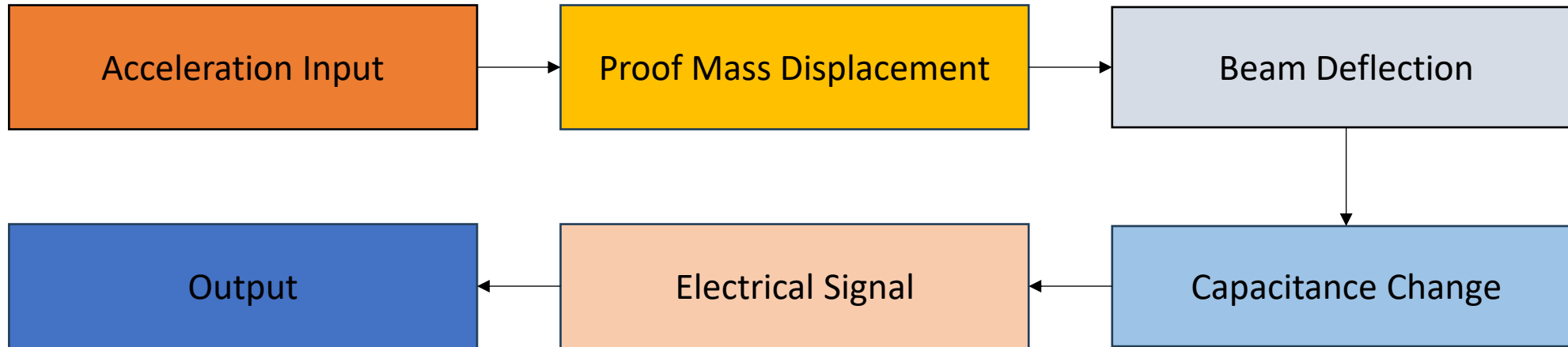
Proposed System

- MEMS-based compact accelerometer design
- Multi-material analysis (Si, GaAs, GaN, SiC)
- High sensitivity and precision
- Optimized beam-proof mass structure
- Suitable for space launch vehicle conditions
- Low power and high reliability





BLOCK DIAGRAM





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Benefits

- Compact and lightweight MEMS structure suitable for space-constrained aerospace systems.
- High sensitivity and accurate acceleration measurement under extreme launch conditions.
- Low power consumption, making it suitable for space and satellite applications.
- Improved mechanical stability and reliability under vibration and shock loads.
- Easy integration with navigation, guidance, and structural monitoring systems.
- Scalable MEMS fabrication enables cost-effective and mass production capability.



Simulation & Result : Displacement

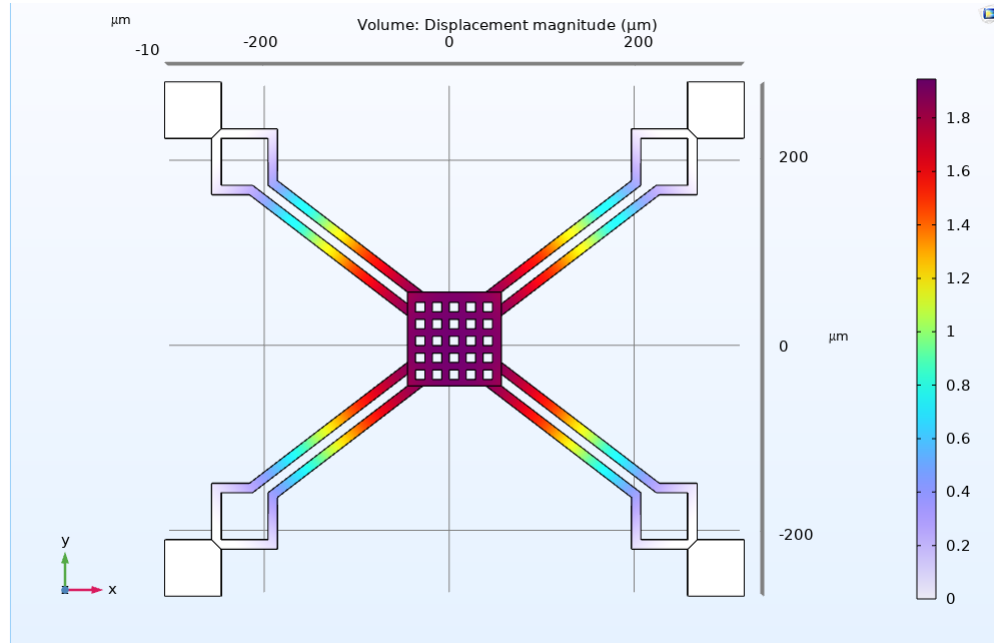


Fig1: Silicon Material

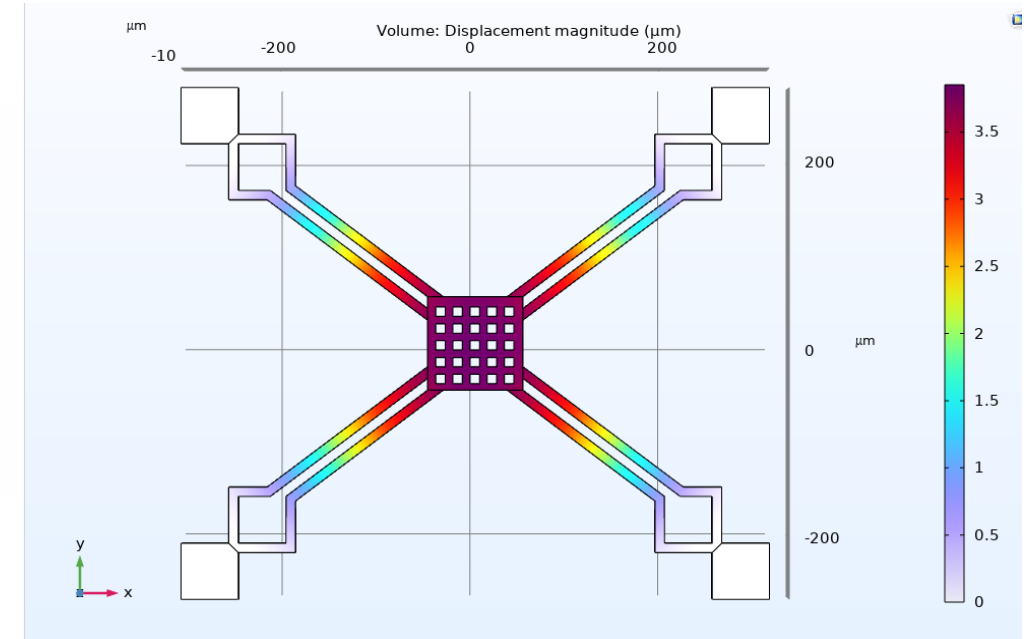


Fig2: Gallium arsenide



Simulation & Result : Displacement

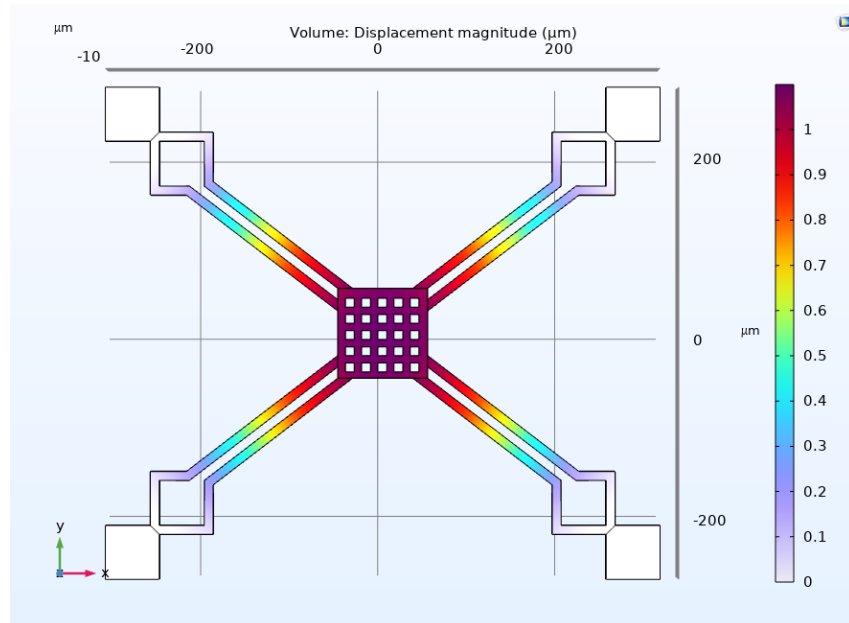


Fig3: Gallium Nitride (GaN)

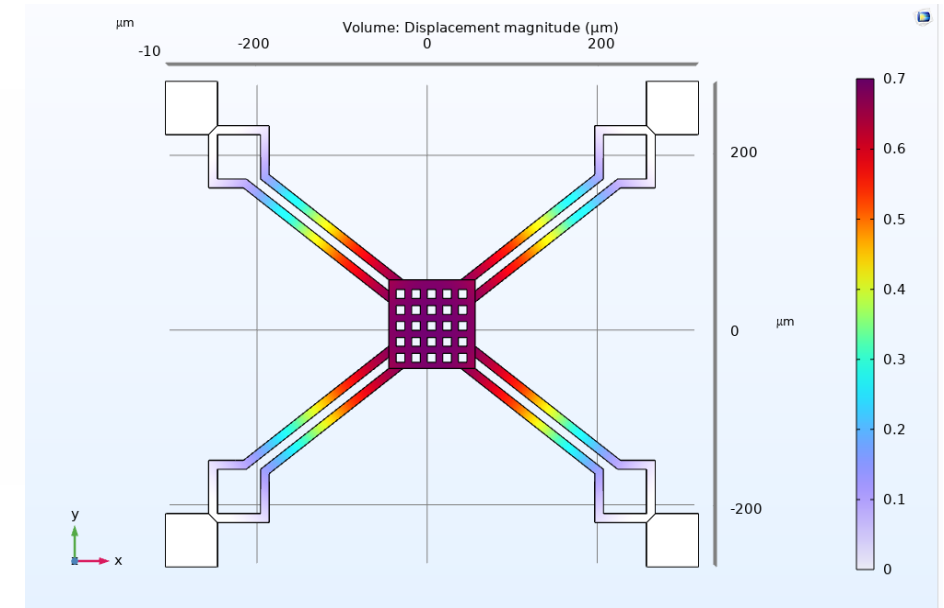


Fig3: Silicon Carbide – SiC (6H)



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Simulation&Result: Strain Analysis

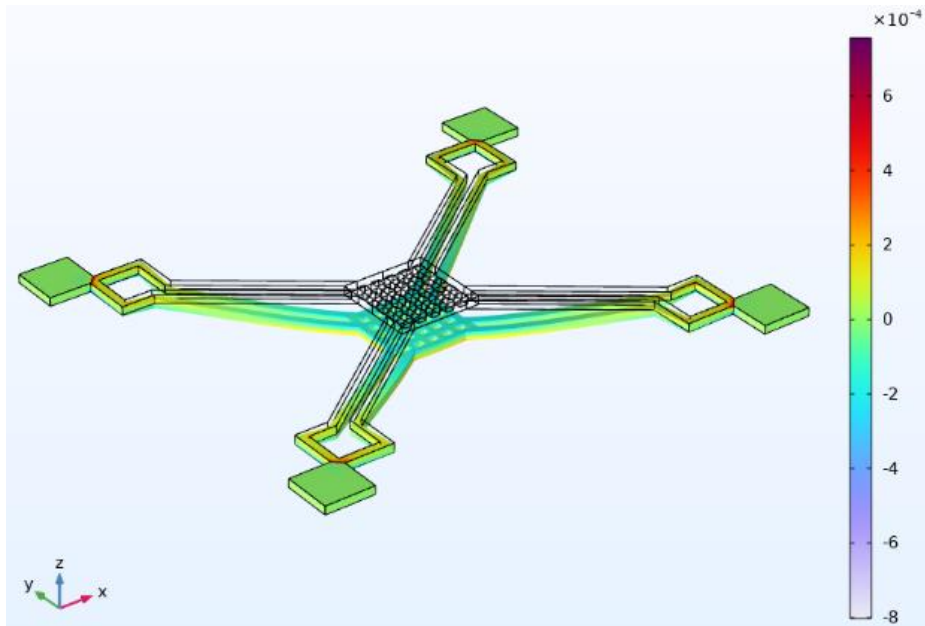


Fig1: Silicon Material

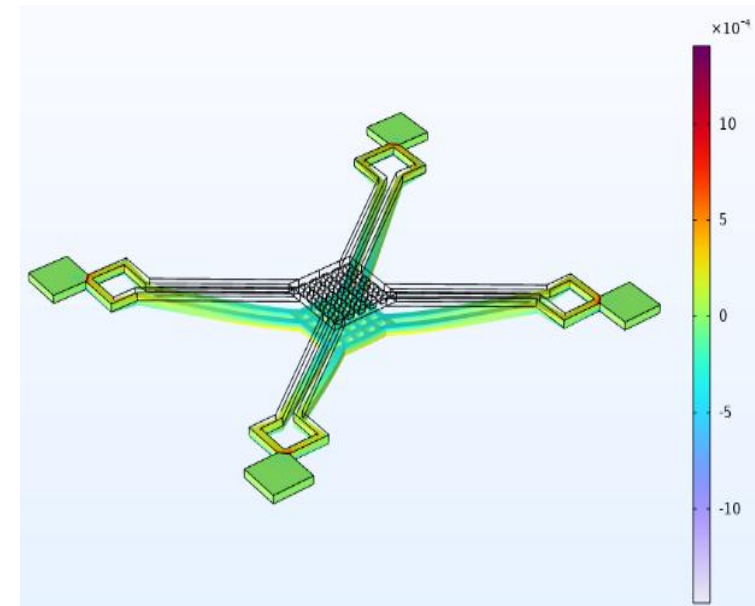


Fig2: Gallium arsenide



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Simulation&Result: Strain Analysis

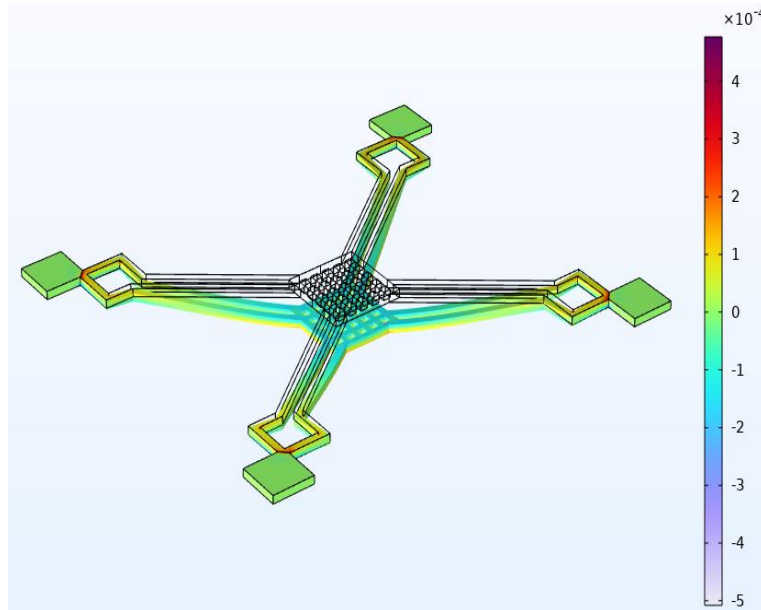


Fig3: Gallium Nitride (GaN)

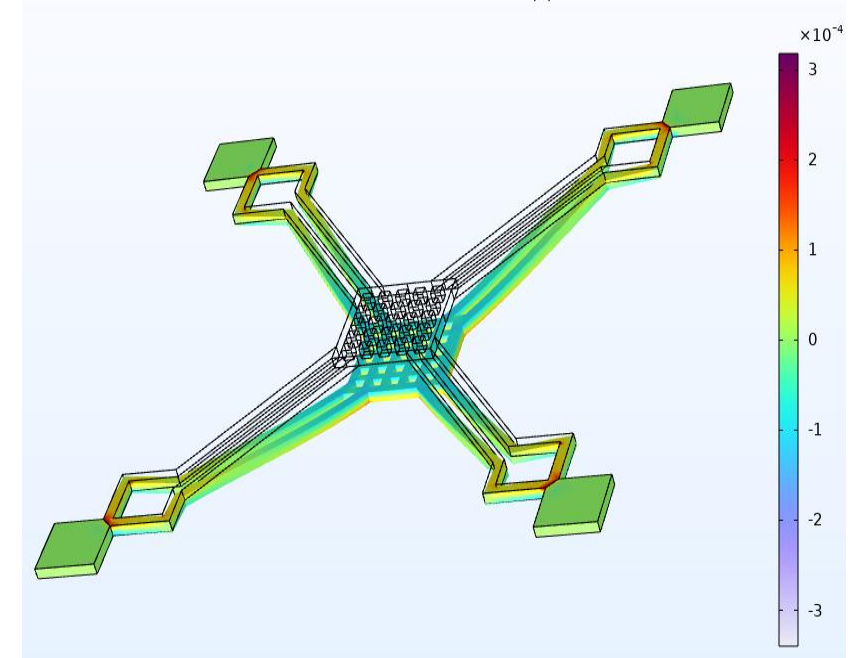


Fig3: Silicon Carbide – SiC (6H)



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Conclusion

A MEMS-based accelerometer for space launch vehicle applications has been successfully designed and analyzed.

The proposed symmetric proof-mass structure provides high sensitivity, mechanical stability, and reliable performance under extreme launch conditions. The simulation results confirm its suitability for aerospace navigation and structural monitoring applications.